

PRUEBA PRÁCTICA 1

Sistemas de Ayuda a la Decisión

Grupo : GO-1-1

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Instrucciones:

- Lee cuidadosamente todas las preguntas de enunciado.
- Justifica todas las respuestas.
- **Incluye los pantallazos solicitados.**
- **Entrega el documento en formato PDF.**

Enunciado

Lee el enunciado que se te ha proporcionado. Este documento es el que tendrás que entregar una vez rellenado en formato PDF.

Tarea 1. Análisis de la variable objetivo

- Número de instancias de cada clase

Hay 44993 instancias para la clase de 0 impagos y 5007 instancias para la clase de 1 impago.

- Porcentaje de cada clase. Recuerda, sabemos que la tasa de impagos se ha incrementado y ahora es mayor del 1\% (no hay errores) ¿En cuanto se ha incrementado?

El 90.0% de las instancias totales pertenecen a la clase de 0 impagos. En cambio, el 10.0% pertenece a la de 1 impago.

- ¿Está desbalanceado el dataset?

Si, está desbalanceado, ya que la diferencia de instancias entre las dos clases de impago es muy grande. Debería ser cercano al 50%.

- ¿Por qué Accuracy no es adecuada?

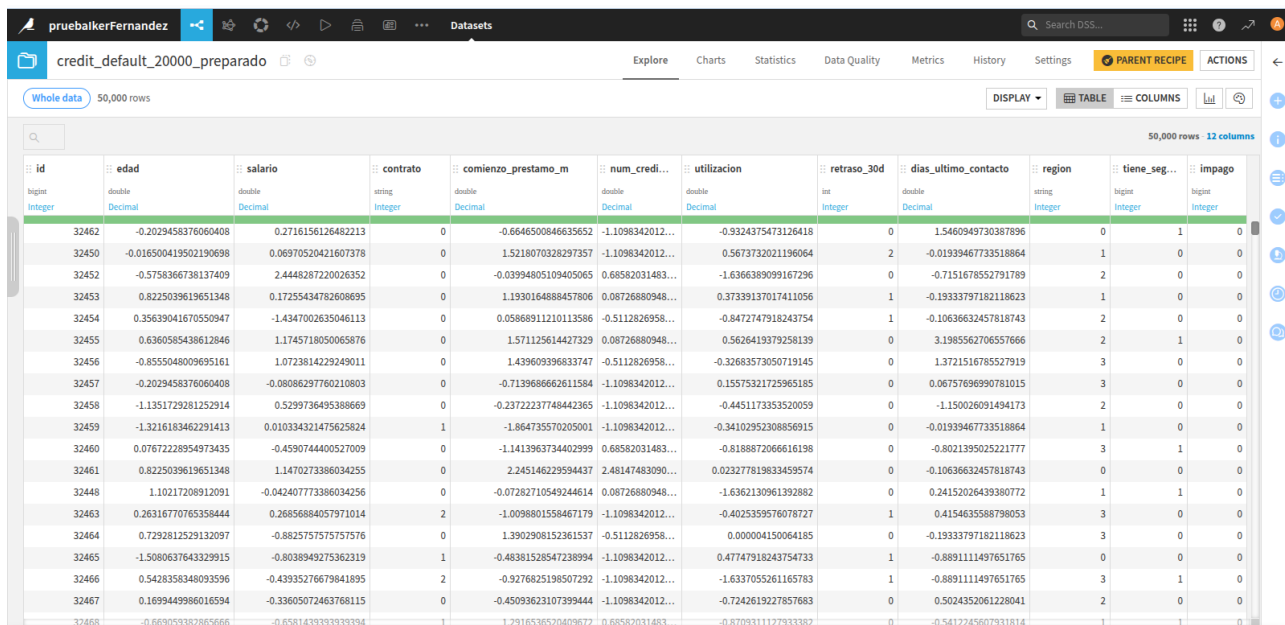
Debido a la gran diferencia entre cantidades de instancias entre las dos clases, el modelo siempre va a tender a predecir que hay 0 impagos. Por lo que, la Accuracy será alta respecto a la Precision.

Tarea 2. Preprocesado

- Tratamiento de valores faltantes
- Conversión de variables categóricas
- Escalado de variables

Pantallazo 1:

No se ha escalado la variable id, ya que representa un identificador. Por otro lado, tampoco se ha escalado la variable retraso_30d ya que sus clases numéricas tienen un intervalo muy pequeño [0,6]

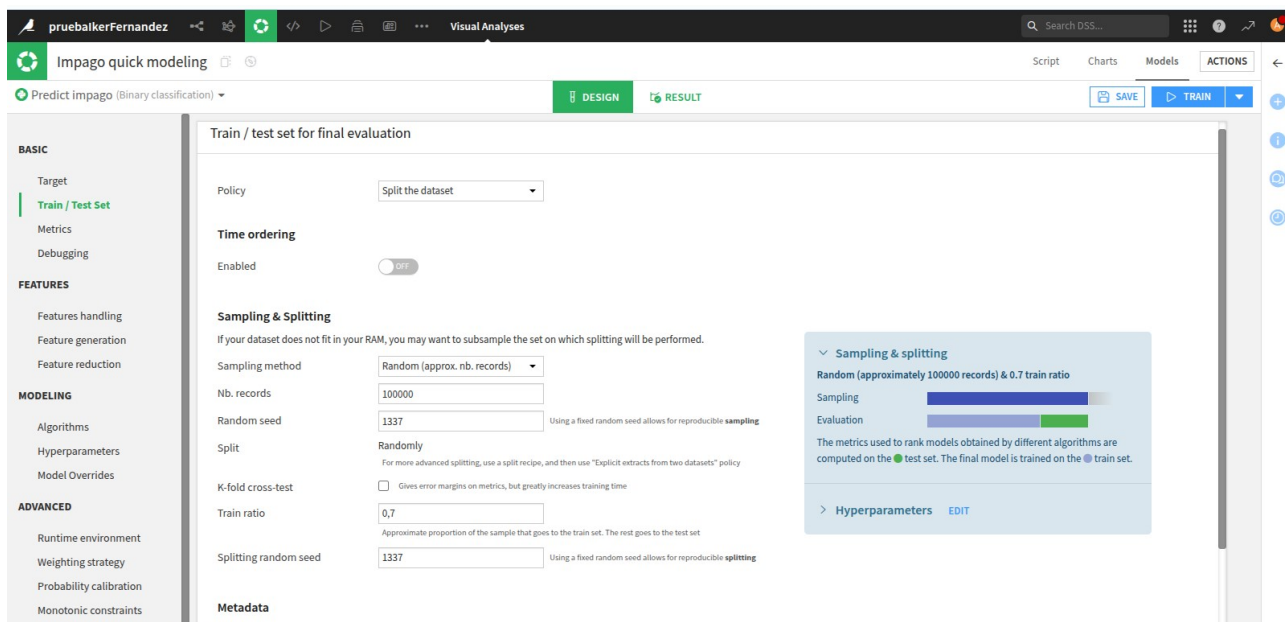


id	edad	salario	contrato	comienzo_prestamo_m	num_cred...	utilizacion	retraso_30d	dias_ultimo_contacto	region	tiene_seg...	impago
bigint Integer	double Decimal	double Decimal	string Integer	double Decimal	double Decimal	double Decimal	int Integer	double Decimal	string Integer	bigint Integer	bigint Integer
32462	-0.2029458376060408	0.2716156126482213	0	-0.6646500846635652	-1.1098342012...	-0.9324375473126418	0	1.5460949730387896	0	1	0
32450	-0.016500419502190698	0.06970520421607378	0	1.5218070328297357	-1.1098342012...	0.56737320221196064	2	-0.01939467733518864	1	0	0
32452	-0.5758366738137409	2.4448287220026352	0	-0.03994805109405065	0.68582031483...	-1.6366389099167296	0	-0.7151678552791789	2	0	0
32453	0.8225039619651348	0.17255434782608695	0	1.1930164889457806	0.08726880948...	0.37339137017411056	1	-0.19333797182118623	1	0	0
32454	0.35639041670550947	-1.4347002635046113	0	0.05868911210113586	-0.5112826958...	-0.8472747918243754	1	-0.10636632457818743	2	0	0
32455	0.6360585438612846	1.1745718050065876	0	1.571125614427329	0.08726880948...	0.5626419379258139	0	3.1985562706557666	2	1	0
32456	-0.8555048009695161	1.0723814229249011	0	1.439609396833747	-0.5112826958...	-0.32683573050719145	0	1.3721516785527919	3	0	0
32457	-0.2029458376060408	-0.08086297760210803	0	-0.7139686662611584	-1.1098342012...	0.15575321725965185	0	0.06757696990781015	3	0	0
32458	-1.1351729281252914	0.5299736495388669	0	-0.23722237748442365	-1.1098342012...	-0.4451173353520059	0	-1.150026091494173	2	0	0
32459	-1.3216183462291413	0.010334321475625624	1	-1.864735570205001	-1.1098342012...	-0.34102952308856915	0	-0.01939467733518864	1	0	0
32460	0.07672228954973435	-0.4590744400527009	0	-1.1413963734402999	0.68582031483...	-0.8188872066616198	0	-0.8021395025221777	3	1	0
32461	0.8225039619651348	1.1470273386034255	0	2.24514622959437	2.48147483090...	0.023277819833459574	0	-0.10636632457818743	0	0	0
32448	1.10217208912091	-0.042407773386034256	0	-0.07282710549244614	0.08726880948...	-1.6362130961392882	0	0.24152026439380772	1	1	0
32463	0.26316770765358444	0.26856884057971014	2	-1.0098801558467179	-1.1098342012...	-0.4025359576078727	1	0.4154635588798053	3	0	0
32464	0.7292812529132097	-0.8825757575757576	0	1.3902908152361537	-0.5112826958...	0.000004150064185	0	-0.19333797182118623	3	0	0
32465	-1.5080637643322915	-0.8038949275362319	1	-0.48381528547238994	-1.1098342012...	0.47747918243754733	1	-0.889111497651765	0	0	0
32466	0.5428358348093596	-0.43935276679841895	2	-0.9276825198507292	-1.1098342012...	-1.6337055261165783	1	-0.889111497651765	3	1	0
32467	0.1699449986016594	-0.33605072463768115	0	-0.4509362310739944	-1.1098342012...	-0.7242619227857683	0	0.5024352061228041	2	0	0
32468	-0.66909382865666	-0.6581433333333334	1	1.2916536520409672	0.68582031483...	-0.870931117933382	0	-0.5412245607931814	1	1	0

Tarea 3. División Train/Test

- Train: 70%
- Test: 30%
- Seed: 1337

Pantallazo 2:

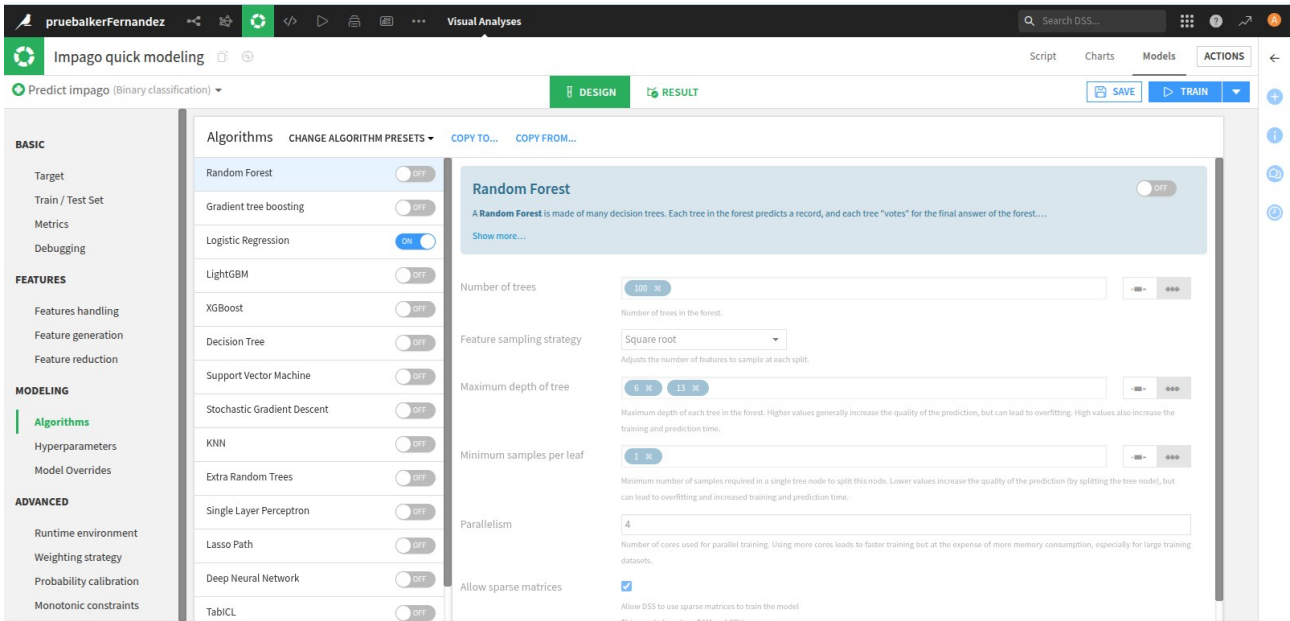


Policy	Split the dataset
Time ordering	Enabled
Sampling & Splitting	If your dataset does not fit in your RAM, you may want to subsample the set on which splitting will be performed. Sampling method: Random (approx. nb. records) Nb. records: 100000 Random seed: 1337 Split: Randomly K-fold cross-test: ☐ Gives error margins on metrics, but greatly increases training time Train ratio: 0.7 Splitting random seed: 1337
Metadata	

Tarea 4. Modelo

Algoritmo: Logistic Regression

Pantallazo 3:



The screenshot shows the 'Impago quick modeling' interface in the Visual Analyses tool. The 'Algorithms' panel on the left lists various models, with 'Random Forest' selected. The 'DESIGN' tab is active, showing the configuration for the Random Forest model. The 'Number of trees' is set to 100, 'Feature sampling strategy' is 'Square root', 'Maximum depth of tree' is 6, 'Minimum samples per leaf' is 1, and 'Parallelism' is set to 4. The 'Allow sparse matrices' checkbox is checked.

Tarea 5. Evaluación

Métrica 15078

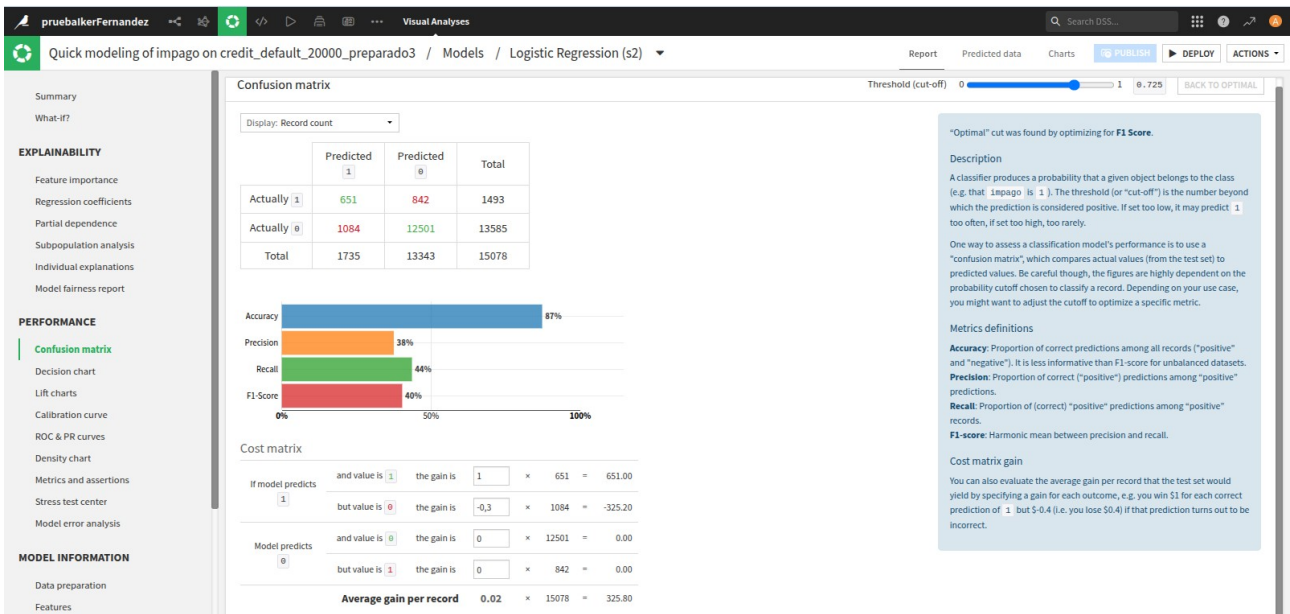
Accuracy 87%

Precision 38%

Recall 44%

F1-score 40%

Pantallazo 4:



The screenshot shows the 'Quick modeling of impago on credit_default_20000_preparado3' interface. The 'Models' tab is active, showing the 'Logistic Regression (s2)' model. The 'Confusion matrix' is displayed, showing the following results:

	Predicted 1	Predicted 0	Total
Actually 1	651	842	1493
Actually 0	1084	12501	13585
Total	1735	13343	15078

The 'PERFORMANCE' section shows the following metrics:

- Accuracy: 87%
- Precision: 38%
- Recall: 44%
- F1-Score: 40%

The 'Cost matrix' section shows the following results:

If model predicts	and value is	the gain is			
1	1	1	×	651	= 651.00
1	0	-0.3	×	1084	= -325.20
0	0	0	×	12501	= 0.00
0	1	0	×	842	= 0.00
Average gain per record				0.02	× 15078 = 325.80

The 'EXPLAINABILITY' section shows the following metrics:

- Feature importance
- Regression coefficients
- Partial dependence
- Subpopulation analysis
- Individual explanations
- Model fairness report

The 'MODEL INFORMATION' section shows the following metrics:

- Data preparation
- Features

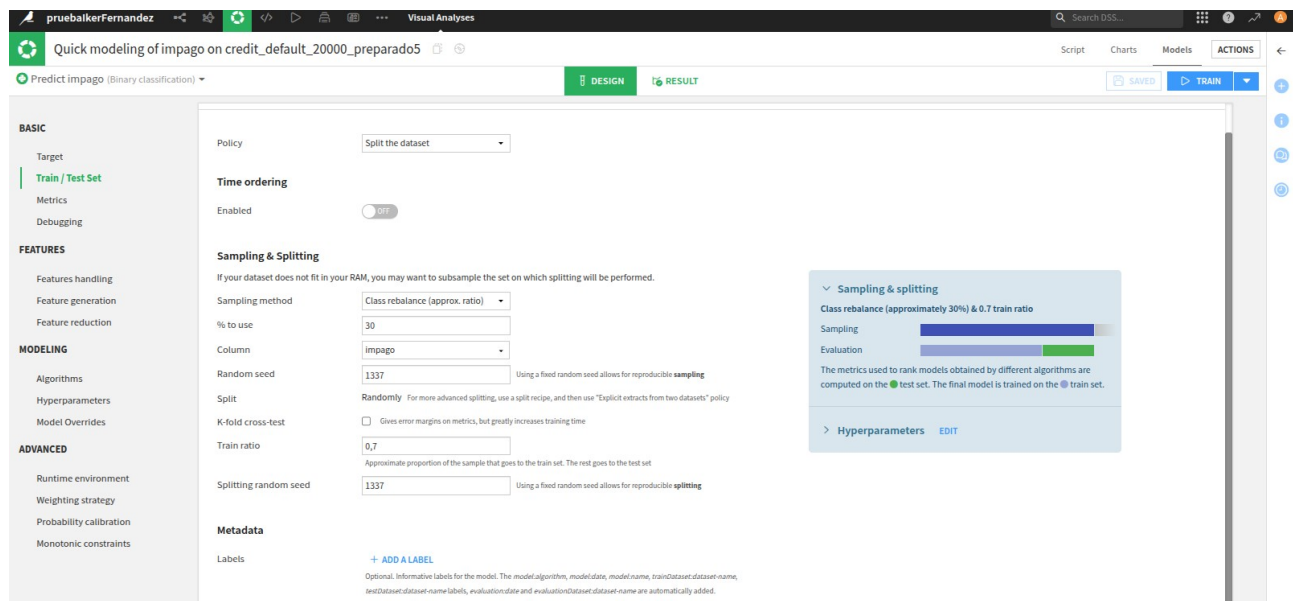
The 'Description' section provides a detailed explanation of the model's performance and the cost matrix.

Tarea 6. Balanceo

Describe el método utilizado:

El método de balanceo se basa en disminuir la muestra de aquella clase que aparece en muchas instancias para aumentar la probabilidad de aparición de aquella que aparece menos, para así mejorar las predicciones. Es decir, aumenta el número de instancias con impago 1 en nuestro ejemplo.

Pantallazo 5:



Visual Analyses

Quick modeling of impago on credit_default_20000_preparado5

Predict impago (Binary classification)

BASIC

Target: Train / Test Set

Metrics

Debugging

FEATURES

Features handling

Feature generation

Feature reduction

MODELING

Algorithms

Hyperparameters

Model Overrides

ADVANCED

Runtime environment

Weighting strategy

Probability calibration

Monotonic constraints

DESIGN

Policy: Split the dataset

Time ordering

Enabled: Off

Sampling & Splitting

If your dataset does not fit in your RAM, you may want to subsample the set on which splitting will be performed.

Sampling method: Class rebalance (approx. ratio)

% to use: 30

Column: impago

Random seed: 1337

Split: Randomly

K-fold cross-test: ☐ Gives error margins on metrics, but greatly increases training time

Train ratio: 0,7

Splitting random seed: 1337

Metadata

Labels: + ADD A LABEL

Optional. Informative labels for the model. The model.algorithm, model.data, model.name, trainDataset.dataset-name, testDataset.dataset-name labels, evaluation.date and evaluationDataset.dataset-name are automatically added.

SAMPLING & SPLITTING

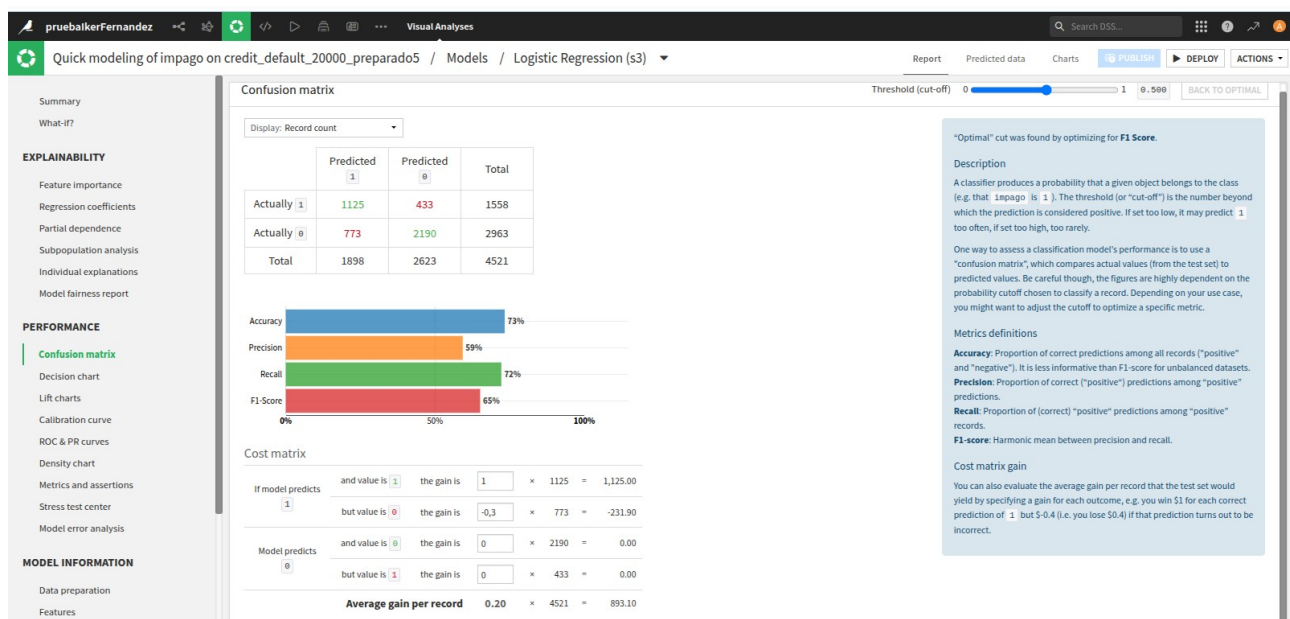
Class rebalance (approximately 30%) & 0.7 train ratio

Sampling: 30%

Evaluation: 0.7

The metrics used to rank models obtained by different algorithms are computed on the test set. The final model is trained on the train set.

> Hyperparameters EDIT



Visual Analyses

Quick modeling of impago on credit_default_20000_preparado5 / Models / Logistic Regression (s3)

Report Predicted data Charts PUBLISH DEPLOY ACTIONS

Threshold (cut-off) 0 0.500 BACK TO OPTIMAL

EXPLAINABILITY

Feature importance

Regression coefficients

Partial dependence

Subpopulation analysis

Individual explanations

Model fairness report

PERFORMANCE

Confusion matrix

Decision chart

Lift charts

Calibration curve

ROC & PR curves

Density chart

Metrics and assertions

Stress test center

Model error analysis

MODEL INFORMATION

Data preparation

Features

Confusion matrix

Display: Record count

	Predicted 1	Predicted 0	Total
Actually 1	1125	433	1558
Actually 0	773	2190	2963
Total	1898	2623	4521

Accuracy: 73%

Precision: 59%

Recall: 72%

F1 Score: 65%

Cost matrix

Model predicts	and value is	the gain is			
1	1	1	1125	=	1,125.00
	0	-0,3	773	=	-231.90
0	0	0	2190	=	0.00
	1	0	433	=	0.00
Average gain per record			0.20	x	4521 = 893.10

"Optimal" cut was found by optimizing for F1 Score.

Description

A classifier produces a probability that a given object belongs to the class (e.g. that "impago" is 1). The threshold (or "cut-off") is the number beyond which the prediction is considered positive. If set too low, it may predict 1 too often, if set too high, too rarely.

One way to assess a classification model's performance is to use a "confusion matrix", which compares actual values (from the test set) to predicted values. Be careful though, the figures are highly dependent on the probability cutoff chosen to classify a record. Depending on your use case, you might want to adjust the cutoff to optimize a specific metric.

Metrics definitions

Accuracy: Proportion of correct predictions among all records ("positive" and "negative"). It is less informative than F1-score for unbalanced datasets.

Precision: Proportion of correct ("positive") predictions among "positive" predictions.

Recall: Proportion of (correct) "positive" predictions among "positive" records.

F1-score: Harmonic mean between precision and recall.

Cost matrix gain

You can also evaluate the average gain per record that the test set would yield by specifying a gain for each outcome, e.g. you win \$1 for each correct prediction of 1 but \$-0.4 (i.e. you lose \$0.4) if that prediction turns out to be incorrect.

Tarea 7. Análisis final

- ¿Qué modelo tiene mejor Recall?

Tiene mejor Recall el segundo, porque ha aumentado la muestra de impagos 1.

- ¿Cuál es el mejor modelo?

Es mejor modelo en segundo, ya que los parámetros de predicción son mejores.

- ¿Por qué es importante Recall en este caso?

Por calcula la proporción de predicciones correctas de 1 impago respecto a la realidad de 1 impago. Por lo tanto, nos permite identificar que ha aumentado la proporción de 1 impago en la muestra respecto a la anterior.